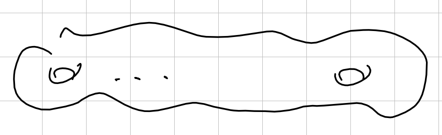
S^n, ~~$S^1\times S^1$~~, $\mathbb{RP}^n$, 

$(n\geq 1)$
$\downarrow$
$\pi_1=0$.

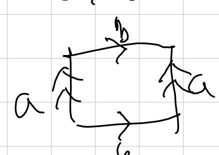
2-dim mfd

$L(p,q)$

$M \# N$. $\wedge$ $\vee$

$S^1\times S^1 / e^{2\pi i p}, e^{2\pi i q}$

$[0,1]^2$



torus.

$\mathbb{Z}\times\mathbb{Z}$

Klein.

Klein.

free product.

~~$\mathbb{Z}*\mathbb{Z}$~~

ababab b.

$\langle a,b \mid aba=b\rangle$.

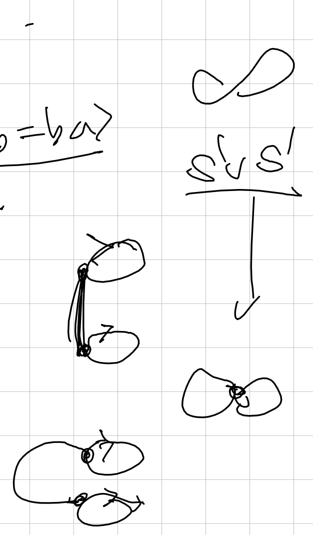
$aba^{-1}b^{-1}=e$.

$\langle a,b \mid ab=ba\rangle$

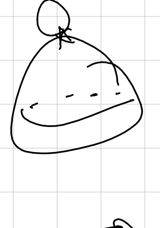
$\mathbb{Z}\times\mathbb{Z}$

$ab=ba \to$ grup

$S^1\vee S^1$



$S^1\vee S^2$



$S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}$

$T: \mathbb{Z}\times\mathbb{Z}$

$T\setminus\circ: \mathbb{Z}*\mathbb{Z}$

$K\setminus\circ: \mathbb{Z}*\mathbb{Z}$

theorem

Theorem

$S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}$

$S^1 \leftrightarrow \mathbb{Z}$

$S^3 : 0$
$\downarrow$
$\mathbb{R}^3 : 0$.

M: n-dim with $(n\geq 3)$

then $\pi_1(M)\cong\pi_1(M\setminus\circ)$.

$\cong\pi_1(M\setminus 2pt)$. $\pi_1(S^n)=0$ $(n\geq 2)$.

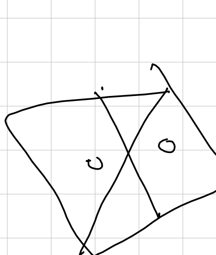
$\cong\pi_1(M\setminus\{n\})$.

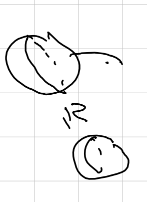
$\pi_1(\mathbb{RP}^n)=\mathbb{Z}_2$ $(n\geq 2)$.

tree

$S^1\vee S^1\vee \mathbb{R}$

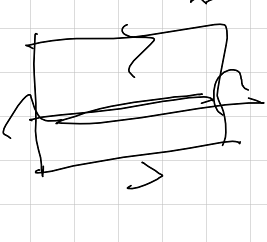
$\mathbb{Z}*\mathbb{Z}$

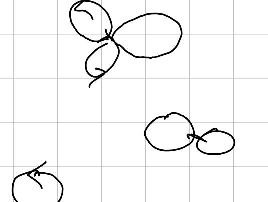




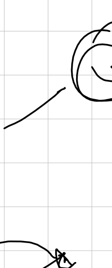
$: S^1\vee S^1\vee S^1$

$\mathbb{Z}*\mathbb{Z}*\mathbb{Z}$

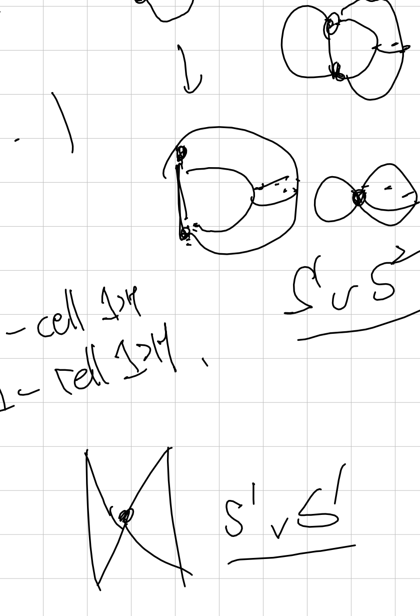


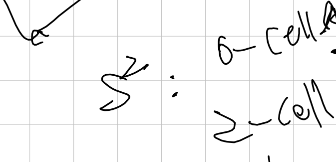


Exemple.



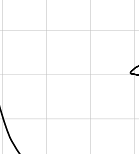
$S^2/N=S$ N

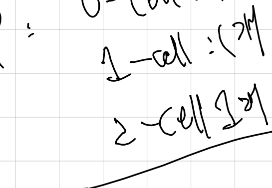




$\to\pi_1(\cdots)$

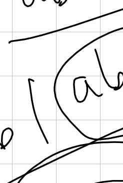
$S^1\vee S^2$





S^1: 0-cell 1개
1-cell 1개

S^2: 0-cell 1개
2-cell 1개



$S^1\vee S^1$

$\mathbb{RP}^2$: 0-cell : 1개
1-cell : 1개
2-cell 1개

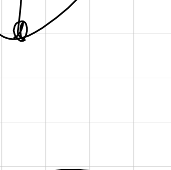
$aba^{-1}b^{-1}=0$.

$\langle a,b \mid ab=ba\rangle$

$(\mathbb{Z}*\mathbb{Z})/\alpha$

$\mathbb{RP}^2 : \mathbb{Z}_2$ $\langle a \mid a^2\rangle$

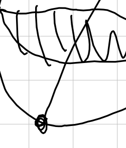
$a^2=1$

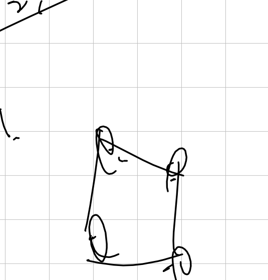


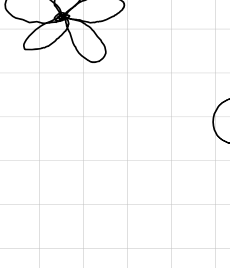
$\mathbb{RP}^3$: 0-cell 1개
1-cell 1개
2-cell 1개
~~3-cell 1개~~

~~$\pi_1(\ldots)$~~

$\pi_1(\mathbb{RP}^3)=\pi_1(\mathbb{RP}^2)=\mathbb{Z}_2$.

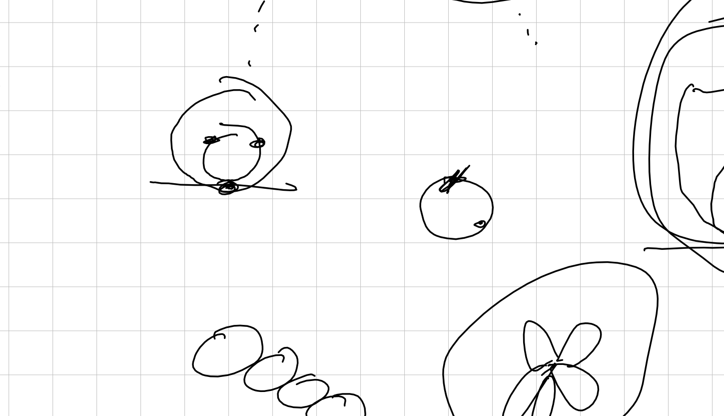




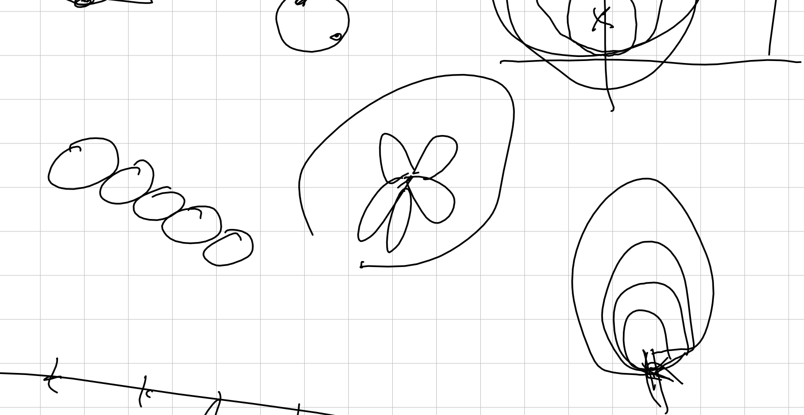


$\leftarrow$ $x^2+y^2+z^2=1$
$\cup$ $x^2+y^2\leq 1$
$\cup$ $(0,0,z)$ $-1\leq z\leq 1$

$(\frac{1}{2},\frac{1}{2},\ldots)$

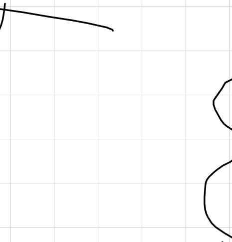


$\mathbb{R}/\mathbb{Z}$



S^1

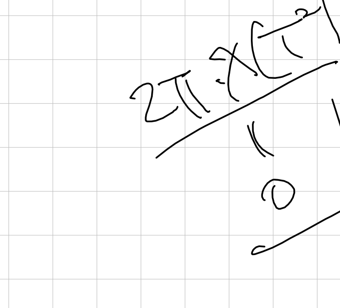
?



$z=x^4+y^4$

$\downarrow$ exp

$K_1=f(4,2)$
$K_2=f(4)$



$\theta=0$

$\frac{1}{R^2+z}$.

$\int\frac{R\,dz}{y}$ $\int\frac{1}{z}dz$

$2\pi i$

$2\pi i\cdot\chi(\ldots)$

$\|$

0